

Dr. Anurag Bajpai

Group Leader, Artificial Intelligence for Materials Science

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Summary of Qualifications

- Materials scientist working at the interface of physical metallurgy, autonomous materials design and sustainable metallurgy, with emphasis on bulk metallic systems, process–microstructure–property relationships, thermodynamic modeling, experimental validation and machine-guided alloy/process design.
- Group Leader of the Artificial Intelligence for Materials Science group at the Max Planck Institute for Sustainable Materials, contributing to the scientific coordination of an international research team involving 2 Master's students, 1 PhD researcher, and 3 postdoctoral researchers.
- Experience developing international collaborative research proposals and PI-level AI-based materials design work packages, including submitted Horizon Europe proposals and a Korea–Germany bilateral research proposal.
- Research portfolio spans hydrogen-based reduction for green steelmaking, scrap-tolerant alloy design, multicomponent metallic glasses, high-entropy alloys, and sustainable materials processing.
- Experienced in building and leading collaborations across computational modeling, alloy processing, microscopy, thermodynamics, and data-driven materials discovery.
- Skilled in mentoring early-career researchers and integrating machine learning, physical metallurgy, microstructure analysis, and experimental validation into coherent research workflows.

Areas of Specialization

- Physical and mechanical metallurgy of bulk metallic systems
- Autonomous and physics-informed materials design
- Machine learning for microstructure-sensitive alloy design and process–microstructure–property relationships
- Sustainable metallurgy and scrap-tolerant alloy design
- Hydrogen-based reduction and green steelmaking
- Bulk metallic glasses, high-entropy and compositionally complex alloys
- Critical-material recovery and circular metallurgy

Technical Skills

Computational materials science: CALPHAD-informed thermodynamic modeling, atomistic simulation, molecular dynamics, composition–processing–property modeling, physics-informed machine learning.

Machine learning and data science: supervised and unsupervised learning, active learning, transfer learning, generative modeling, uncertainty quantification, symbolic regression, image segmentation, feature engineering, model interpretation.

Programming and software: Python, NumPy, pandas, SciPy, scikit-learn, TensorFlow/Keras, MATLAB, LAMMPS, OVITO, Origin.

Materials processing and thermal treatment: vacuum arc melting, mechanical alloying, cryomilling, pulsed laser ablation, heat-treatment design, sample preparation for microstructural (SEM and TEM) and mechanical characterization.

Characterization and microstructure analysis: high-energy synchrotron XRD, SEM, EPMA, DSC, nanoindentation, AFM, Raman spectroscopy, XPS, FTIR, UV–Vis spectroscopy, image-based microstructure quantification.

Education

Indian Institute of Technology Kanpur

M.Tech–PhD Dual Degree, GPA: 8.3/10

Materials Science and Engineering

India

January 2017–August 2023

Ph.D. Thesis: *Compositional Design, Thermal Stability and Mechanical Behavior of Multicomponent Metallic Glasses*

M.Tech Thesis: *From Trash to Treasure - Utilization of Electronic Waste as precursors in Advanced Technologies*

Punjab Engineering College, Chandigarh

Bachelor of Engineering, GPA: 7.7/10

Materials and Metallurgical Engineering

India

2011–2015

Professional Experience

Max Planck Institute for Sustainable Materials (formerly, MPIE)

Germany

Group Leader, Artificial Intelligence for Materials Science

Alexander von Humboldt Postdoctoral Fellow

June 2024–present

○ Scientific host: Prof. Dierk Raabe

- Lead research activities in physics-informed and uncertainty-aware machine learning for autonomous materials design, with emphasis on sustainable metallurgy, bulk metallic systems, and process–microstructure–property relationships.
- Integrate machine learning, thermodynamic constraints, microstructure analysis, and experimental validation into coherent autonomous metallurgy workflows.
- Scientifically coordinate research activities in the group, including technical mentoring, research planning, framework development, and research manuscript coordination.

Max Planck Institute for Iron Research (MPIE)

Germany

Max Planck Gesellschaft Postdoctoral Fellow

March 2023–May 2024

○ Scientific host: Prof. Dierk Raabe

- Developed physics-informed and interpretable machine-learning models for metallic materials design, including applications to hydrogen-based reduction, multicomponent metallic glasses, and sustainable alloy discovery.
- Integrated thermodynamic reasoning, experimental metallurgy, atomistic insights, and data-driven modeling to study deformation, relaxation, crystallization, and process-dependent property evolution in complex metallic systems.

Tata Motors

India

Technical Manager, Vehicle Layout Operations

September 2015–October 2016

Selected Research Experience

Autonomous bulk materials design and process–microstructure–property modeling

(MPISusMat, Present):

- Developed physics-informed and uncertainty-aware machine-learning frameworks for autonomous materials design, with applications to multicomponent metallic glasses, sustainable alloys, and hydrogen-based metallurgical processes.
- Introduced attention-enhanced variational learning for inverse design of exceptionally hard multicomponent bulk metallic glasses, combining latent-space learning, physical descriptors, and targeted property optimization.
- Developed physics-constrained constitutive learning strategies to identify rate-limiting timescales in hydrogen-based direct reduction, supporting interpretable process modeling for green steelmaking.

Scrap-tolerant structural alloys and circular metallurgy

(MPISusMat, Present):

- Develop active-learning workflows for the design of high-Cu-tolerant advanced steels motivated by scrap-based alloy compositions and circular metallurgy.
- Integrate thermodynamic modeling, composition descriptors, microstructure-sensitive feature extraction, and uncertainty-aware property prediction to identify alloying strategies compatible with elevated residual impurity contents.
- Develop microscopy-based image segmentation and feature-extraction pipelines to quantify microstructural descriptors for process–microstructure–property correlation.
- Establish scrap-tolerant alloy design as a route toward sustainable structural metals with improved raw-material efficiency and reduced dependence on primary alloying resources.

Physical/mechanical metallurgy of bulk metallic glasses and compositionally complex alloys

(IITK, 2017-2023):

- Developed machine-learning and thermodynamic-design strategies for multicomponent metallic glasses, including Mendeleev-number-based composition mapping and high-throughput glass-forming ability prediction.
- Combined calorimetry, nanoindentation, microstructure characterization, high-energy X-ray diffraction, deformation modeling, and molecular dynamics to study crystallization, structural relaxation, serrated flow, and time-dependent deformation in Cu–Zr-based metallic glasses.
- Established composition–structure–property relationships linking compositional complexity, free-volume sensitivity, deformation-zone connectivity, and intermittent plasticity in metallic glasses.

Sustainable metallurgy and resource recovery

(IITK, 2017-2022):

- Developed cryomilling-based routes to separate and recover metallic and polymeric fractions from electronic waste for sustainable materials processing.
- Demonstrated downstream utilization of recovered components, including electrochemical CO₂ reduction, graphene synthesis from polymeric waste fractions, and rare-earth recovery using EDTA-functionalized chitosan.

Research Publications

1. **Anurag Bajpai**, Jaemin Wang, Barak Ratzker, Bilgehan Murat Sesen, Florian Kark, Dierk Raabe; *Attention-enhanced variational learning for physically informed discovery of exceptionally hard multicomponent bulk metallic glasses*. **Nature Communications**, Volume 17, 2026. DOI: 10.1038/s41467-026-73008-0. *Selected as Feature Article in Materials Science and Chemistry theme.*

2. **Anurag Bajpai**, Jaemin Wang, Barak Ratzker, Rebeca Miyar, Hendrik Holz, Nilesh P. Gurao, Krishanu Biswas, Dierk Raabe; *Coupled influence of structural relaxation and compositional complexity on deformation behavior and intermittent plasticity in Cu–Zr-based metallic glasses*. **Acta Materialia**, Volume 315, 2026. DOI: 10.1016/j.actamat.2026.122394.
3. **Anurag Bajpai**, Barak Ratzker, Pasquale Cavaliere, Dierk Raabe; *Physics-constrained constitutive learning of rate-limiting timescales for efficient hydrogen-based direct reduction for green steelmaking*. **Advanced Science**, Volume 14, 2026. DOI: 10.1002/advs.75498.
4. **Anurag Bajpai**, Jaemin Wang, Dierk Raabe; *Compositional complexity buffers free-volume sensitivity and serrated flow in metallic glasses*. **npj Computational Materials**, Volume 12, 2026. DOI: 10.1038/s41524-025-01933-7.
5. **Anurag Bajpai**, Jaemin Wang, Dierk Raabe; *Deformation zone connectivity governs time-dependent load relaxation in Cu–Zr-based metallic glasses*. **Computational Materials Science**, Volume 269, 2026. DOI: 10.1016/j.commatsci.2026.114716. *Selected as Editor's Choice Article of the Month for April 2026*.
6. Ziyuan Rao*, **Anurag Bajpai*** (*equal contribution), Hongbin Zhang; *Active learning strategies for design of sustainable alloys*. **Philosophical Transactions of the Royal Society–A**, Volume 382, 2024. DOI: 10.1098/rsta.2023.0242.
7. **Anurag Bajpai** and Krishanu Biswas; *Thermal Stability of newly developed Cu–Zr–Ag–Ti–Ni Multicomponent Bulk Metallic Glass*. **Materials Chemistry and Physics**, Volume 307, 2023. DOI: 10.1016/j.matchemphys.2023.128092.
8. **Anurag Bajpai**, Jatin Bhatt, Krishanu Biswas, and Nilesh P. Gurao; *Accelerated design of Multicomponent Metallic Glasses using Machine Learning*. **Journal of Materials Research**, Volume 37, 2022. DOI: 10.1557/s43578-022-00659-2. *Selected as Cover Article for Volume 37, Issue 15*.
9. **Anurag Bajpai**, Jatin Bhatt, Krishanu Biswas, and Nilesh P. Gurao; *A new approach to design Multicomponent Metallic Glasses using Mendeleev Number*. **Philosophical Magazine**, 2022. DOI: 10.1080/14786435.2022.2121868.
10. **Anurag Bajpai**, Partha Kumbhakar, Chandra Sekhar Tiwary, and Krishanu Biswas; *Conducting graphene synthesis from electronic waste*. **ACS Sustainable Chemistry and Engineering**, Volume 42, 2021. DOI: 10.1021/acssuschemeng.1c03817. *Selected as Cover Article for Volume 9, Number 42*.
11. Nidhi Sharma*, **Anurag Bajpai*** (*equal contribution), Chandra Sekhar Tiwary, and Krishanu Biswas; *Green Route for Beneficiation of Metallic Materials from Electronic Waste for Selective Reduction of CO₂*. **ACS Sustainable Chemistry and Engineering**, Volume 32, 2020. DOI: 10.1021/acssuschemeng.0c03605. *Selected as Cover Article for Volume 8, Number 32*.
12. **Anurag Bajpai**, Jatin Bhatt, Krishanu Biswas, and Nilesh P. Gurao; *A new perspective to thermodynamical designing of high entropy bulk metallic glasses*. **Physica B: Condensed Matter**, Volume 595, 2020. DOI: 10.1016/j.physb.2020.412350.
13. Shashank Shekhar Mishra, Khushubo Tiwari, **Anurag Bajpai**, Thakur Prasad Yadav, Ram Manohar Yadav, et al.; *Structural Evolution and Tunable Magnetic Response in novel TiZrVCrNiFe–X (X= Mn, MnCo) Alloys: An Experimental and Thermodynamic Study*. **Journal of Alloys and Compounds**, Volume 1055, 2026. DOI: 10.1016/j.jallcom.2026.186380.
14. Rahul Mitra, **Anurag Bajpai**, Krishanu Biswas; *A critical review on mechanical alloying of high-entropy materials*. **High Entropy Alloys and Materials**, Volume 3, 2025. DOI: 10.1007/s44210-025-00056-4.
15. Shruti Srivastava, **Anurag Bajpai**, and Krishanu Biswas; *Recovery of Rare Earth Elements (Nd, Dy) from spent magnets using EDTA Functionalized Chitosan*. **RSC Sustainability**, Volume 2, 2024. DOI: 10.1039/D3SU00427A.
16. Rahul Mitra, **Anurag Bajpai**, and Krishanu Biswas; *ADASYN-assisted machine learning for phase prediction of high entropy carbides*. **Computational Materials Science**, Volume 223, 2023. DOI: 10.1016/j.commatsci.2023.112142.
17. Rahul Mitra, **Anurag Bajpai**, and Krishanu Biswas; *Machine learning-based approach for phase prediction in High Entropy Borides*. **Ceramics International**, Volume 48, 2022. DOI: 10.1016/j.ceramint.2022.02.218.
18. Bejjipurapu Akhil, **Anurag Bajpai**, and Krishanu Biswas; *Microstructure and Mechanical Properties of the new TiZrHfReAl HCP High Entropy alloy*. **Philosophical Magazine Letters**, 2022. DOI: 10.1080/09500839.2022.2120644.
19. S. S. Mishra, **Anurag Bajpai**, and Krishanu Biswas; *TiVCrNiZrFe_x High entropy alloy: Phase evolution, Magnetic and Mechanical properties*. **Journal of Alloys and Compounds**, Volume 871, 2021. DOI: 10.1016/j.jallcom.2021.159572.
20. S. S. Mishra, **Anurag Bajpai**, and Krishanu Biswas; *Structure-Property Correlation of Cu₂Mn₁Al_{1–x}Ga_x Full-Heusler Alloy*. **Journal of Alloys and Compounds**, Volume 898, 2021. DOI: 10.1016/j.jallcom.2021.162865.
21. Bejjipurapu Akhil, **Anurag Bajpai**, N. P. Gurao, and Krishanu Biswas; *Designing Hexagonal Close Packed High Entropy Alloys using Machine Learning*. **Modelling and Simulation in Materials Science and Engineering**, Volume 29, 2021. DOI: 10.1088/1361-651X/ac2b37.
22. Jitesh Kumar, Saumya Jha, Abheepsit Raturi, **Anurag Bajpai**, Reshma Sonkusare, N.P. Gurao, and Krishanu Biswas; *Novel Alloy Design Concepts Enabling Enhanced Mechanical Properties of High Entropy Alloys*. **Frontiers in Materials**, Volume 9, 2022. DOI: 10.3389/fmats.2022.868721.

Books

Anurag Bajpai, Krishanu Biswas and Arunabh Meshram, *Beneficiation and Management of Technological Waste: Opportunities and Challenges*, CRC Press, Taylor and Francis Group, 2025, ISBN - 9781003619680.

Patents

Anurag Bajpai, Syed Mohammad Musthaq and Krishanu Biswas; Electric Switch Casing using waste printed circuit boards (WPCBs) via Compression Molding (Published, Application No. 202411011268, 2024).

Submitted Research Proposals and Project Leadership

1. **German Principal Investigator**, Korea–Germany Joint Research Program 2026 proposal, Explainable Data-Driven Design of Impurity-Tolerant Recycled Aluminum Alloys; *submitted and under review*.
2. **Work-package Principal Investigator for AI-based materials and processing design**, Horizon Europe proposal **CALLMA**. Responsible for AI-driven materials design concepts linking composition, processing, microstructure, and performance in sustainable alloy development; *submitted and under review*.
3. **Work-package Principal Investigator for causal-inference-based materials design**, Horizon Europe proposal **UNI-FAST**. Responsible for developing AI-assisted causality-driven materials design workflows for sustainable alloy/process optimization and data-driven metallurgical decision-making; *submitted and under review*.

Research Leadership, Mentoring and Professional Service

- **Group Leader**, Artificial Intelligence for Materials Science group at Max Planck Institute for Sustainable Materials, contributing to the scientific coordination of a multidisciplinary team working at the interface of machine learning, physical metallurgy, microstructure characterization, and sustainable alloy design.
- The team includes 2 Master's students, 1 PhD researcher, and 3 postdoctoral researchers; responsibilities include scientific mentoring, technical guidance, research planning, development of computational frameworks, coordination of research manuscripts, and integration of computational and experimental workflows.
- Mentored two M.Tech students during doctoral research; one mentee received the Gold Medal for Best M.Tech Thesis on machine-learning-assisted design of hexagonal close-packed high-entropy alloys.
- Developed and coordinated collaborative research activities involving MPI-SusMat/MPIE, IIT Kanpur, and international collaborators in alloy design, hydrogen-based reduction, metallic glasses, and sustainable metallurgy.
- Experience in setting up applied research directions connecting materials informatics, alloy processing, microstructure analysis, and experimental validation.
- Co-organizer of the *AI-driven Materials Discovery and Design* symposium at the upcoming **MMM12**, Jeju, Korea, 8–13 November 2026.
- Guest Editor, Special Issue *Artificial Intelligence (AI) in Material Design, Processes* (MDPI).
- Active reviewer for **Nature Machine Intelligence**, **Communications Engineering**, **Materials Characterization**, **Journal of Alloys and Compounds**, **Advanced Engineering Materials**, **JOM** and **Transactions of the Indian Institute of Metals**.
- Teaching Assistant for Thermodynamics of Materials, Structure and Characterization of Materials, Process Engineering Laboratory, Mechanical Behavior Laboratory, and Manufacturing Processes Laboratory, at IIT Kanpur.

Conferences and Talks

Invited Talks.....

- 1: **Indian Institute of Technology (BHU) Varanasi**, titled "*Applied Materials Informatics for Alloy Design*" on 25th April 2024.
- 2: **Visvesvaraya National Institute of Technology (VNIT Nagpur)**, titled "*Materials Informatics for Alloy Design - Current Challenges and Opportunities*" on 21st December 2023.
- 3: **Indian Institute of Technology (IIT) Indore**, titled "*Material Informatics based High Entropy Alloy Design*" on 24th November 2023.
- 4: **Indian Institute of Technology (IIT) Kanpur**, organized by Materials Advantage IIT Kanpur Chapter, titled "*Machine Learning in Materials Science*" on 28th September 2023.

Oral/Poster Presentations.....

- 5: Poster Presentation at **Ab initio Description of Iron and Steel (ADIS2023)** organized by Max Planck Society at Rinberg Castle, Tegernsee from 29th October to 3rd November 2023.
- 6: Poster Presentation at **5th Materials Chain International Conference – MCIC 2023** organized by Ruhr-Universität Bochum, Germany on August 31, 2023.
- 7: Poster Presentation at **Research Scholars' Day 2022 (Padarth)** organized by the Department of Materials Science and Engineering, IIT Kanpur on 7th May 2022 (Received **Best Poster Award**).
- 8: Oral Presentation at **Research Scholars' Day 2022 (Padarth)** organized by the Department of Materials Science and Engineering, IIT Kanpur on 7th May 2022.
- 9: Oral Presentation at **Scientific Research Convention 2022** organized by IIT Kanpur and IEEE from 4th to 6th March 2022 (Received **Best Oral Presentation Award**).
- 10: Poster Presentation at **TMS 2022 Annual Meeting** organized by the American Institute of Mining, Metallurgical, and

Petroleum Engineers (AIME) from 3rd to 7th March 2022.

11: Poster Presentation at **Indian Institute of Metals - Annual Technical Meeting (IIM-ATM) 2021** organized by the Indian Institute of Metals and Tata Steel from 13th to 15th November 2021 (Received **Two Best Poster Awards**).

Recognition and Awards

Honours and Fellowships.....

2024: Appointed Group Leader for the **Artificial Intelligence in Materials Science** at the Max Planck Institute for Sustainable Materials (Formerly Max Planck Institute for Iron Research, MPIE, GmbH), Germany.

2024: Awarded the **Alexander von Humboldt Postdoctoral Fellowship** for research at the Max Planck Institute for Sustainable Materials (Formerly Max Planck Institute for Iron Research, MPIE, GmbH), Germany.

2023: Awarded **Research Fellowship** by the **Max-Planck-Gesellschaft (Max Planck Society)** for postdoctoral research at the Max Planck Institute for Iron Research, MPIE, GmbH, Germany.

2016: Recipient of the **MHRD Fellowship** provided by the Government of India to pursue master's and doctoral research.

Awards.....

2026: Article titled "*Attention-enhanced variational learning for physically informed discovery of exceptionally hard multicomponent bulk metallic glasses*" selected for the **Nature Communications Editors' Highlights** under the **Materials science and chemistry** theme.

2026: Article titled "*Deformation zone connectivity governs time-dependent load relaxation in Cu-Zr-based metallic glasses*" selected as **Editor's Choice Article of the Month** for April 2026 of the journal **Computational Materials Science**.

2022: Article titled "*Accelerated Design of Multicomponent Metallic Glasses using Machine Learning*" selected as **Journal Cover** for Volume 37, Issue 15 of the **Journal of Materials Research**.

2022: **Best Poster Award** on Research Scholars' Day 2022 organized by the Department of Materials Science and Engineering, IIT Kanpur, India.

2022: **Best Oral Presentation Award** at Students' Research Convention 2022 organized by the Indian Institute of Technology Kanpur and IEEE.

2021: Article titled "*Conducting Graphene Synthesis from Electronic Waste*" featured as **Journal Cover** for Volume 9, Number 42 of **ACS Sustainable Chemistry and Engineering**.

2021: Won **Two Best Poster Awards** under the *Energy-Environment & Waste Utilization* and *Advances in Material Science & Technology* themes at the **Indian Institute of Metals 75th Annual Technical Meeting**.

2020: Article titled "*A Green Route for Beneficiation of Metallic Materials from Electronic Waste for Selective Reduction of CO₂*" featured as **Journal Cover** for Volume 8, Number 32 of **ACS Sustainable Chemistry and Engineering**.

References

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